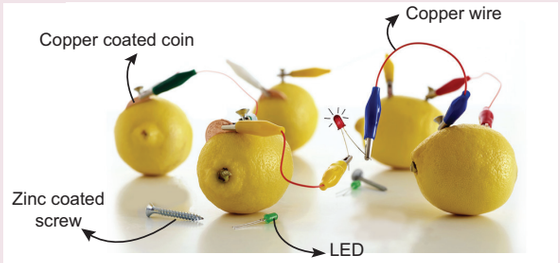
Try Yourself: Construct a Lemon cell as shown in picture.



14.2.3 Uses of Bases

(i) Sodium hydroxide is used in the manufacture of soap.

(ii) Calcium hydroxide is used in white washing of building.

(iii) Magnesium hydroxide is used as a medicine for stomach disorder.

(iv) Ammonium hydroxide is used to remove grease stains from cloths.

14.3 Tests for Acids and Bases

a) Test with a litmus paper:

An acid turns blue litmus paper into red. A base turns red litmus paper into blue.

b) Test with an indicator Phenolphthalein:

In acid medium, phenolphthalein is colourless. In basic medium, phenolphthalein is pink in colour.

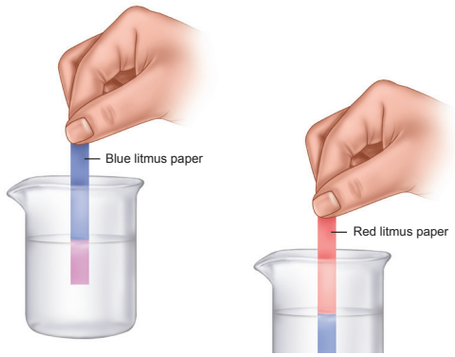


Figure 14.3 Test for acid and base using litmus paper

c) Test with an indicator Methyl orange:

In acid medium, methyl orange is pink in colour. In basic medium, methyl orange is yellow in colour.

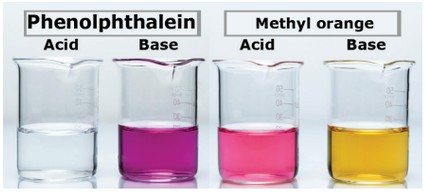


Figure 14.4 Test for acid and base using indicator

Table 14.4 Acid base indicator

Indicator	Colour in acid	Colour in base
Litmus	Blue to Red	Red to Blue
Phenolphthalein	Colourless	Pink
Methyl orange	Pink	Yellow

Activity 3

Collect the following samples from the science laboratory – Hydrochloric acid, Sulphuric acid and Nitric acid, Sodium hydroxide, Potassium hydroxide. Take 2 ml of each solution in a test tube and test with a litmus paper and indicators phenolphthalein and Methyl orange. Tabulate your observations.

Sample Solutions	Litmus Paper		Indicators	
	Blue	Red	Phenolphthalein	Methyl Orange
Hydrochloric acid				
Sulphuric acid				
Nitric acid				
Sodium hydroxide				
Potassium hydroxide				

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